

Design and Implementation of Smart Dry and Wet Waste Segregation System Using Sensors

Triveni Biradar, Sakshi Kadam, Shraddha Upadhye, Sakshi Badave , Dr. Sneha Pokharkar

Department of Electronics and Telecommunication Engineering
MIT Academy of Engineering, Pune, India

Abstract

Improper segregation of municipal solid waste leads to environmental pollution and inefficient recycling processes. This paper proposes a Smart Dry and Wet Waste Segregation System integrated with IoT-based monitoring. The system utilizes an ultrasonic sensor for waste detection and a moisture sensor for classification. An Arduino Uno performs decision-making and controls a servo motor to direct waste into the appropriate bin. A NodeMCU ESP8266 module is used for monitoring bin levels and sending notifications through a Blynk/Firebase platform when bins are full. The proposed system is compact, cost-effective, and suitable for household-level implementation.

Keywords: Waste segregation, Ultrasonic sensor, Moisture sensor, Arduino, Smart bin

INTRODUCTION

The conventional method of waste disposal largely depends on manual segregation or user awareness in separating dry and wet waste. However, manual segregation is time-consuming, inconsistent, and often ignored by households. Lack of proper segregation increases landfill accumulation and contributes to soil and groundwater contamination. Therefore, segregation at the primary level, i.e., at the household itself, is essential for effective waste management. According to a sanitation survey called “Swachh Survekshan - 2016” conducted by the ministry of urban development under the swachh bharat mission, it was found that about 50% people in India face the problem of improper waste collection and management. According to centre of science and environment, innovative disposal and recycling methods must be introduced instead of landfill sites.

To address this issue, a compact and cost-effective **Smart Dry and Wet Waste Segregation System** is proposed. The system automates the segregation process using sensor-based detection. An ultrasonic sensor is used to detect the presence of waste, and a moisture sensor determines whether the waste is wet or dry based on its moisture content. Depending on the sensor output, a servo-controlled mechanism directs the waste into the appropriate bin. This approach minimizes human intervention and ensures accurate classification at the source level.

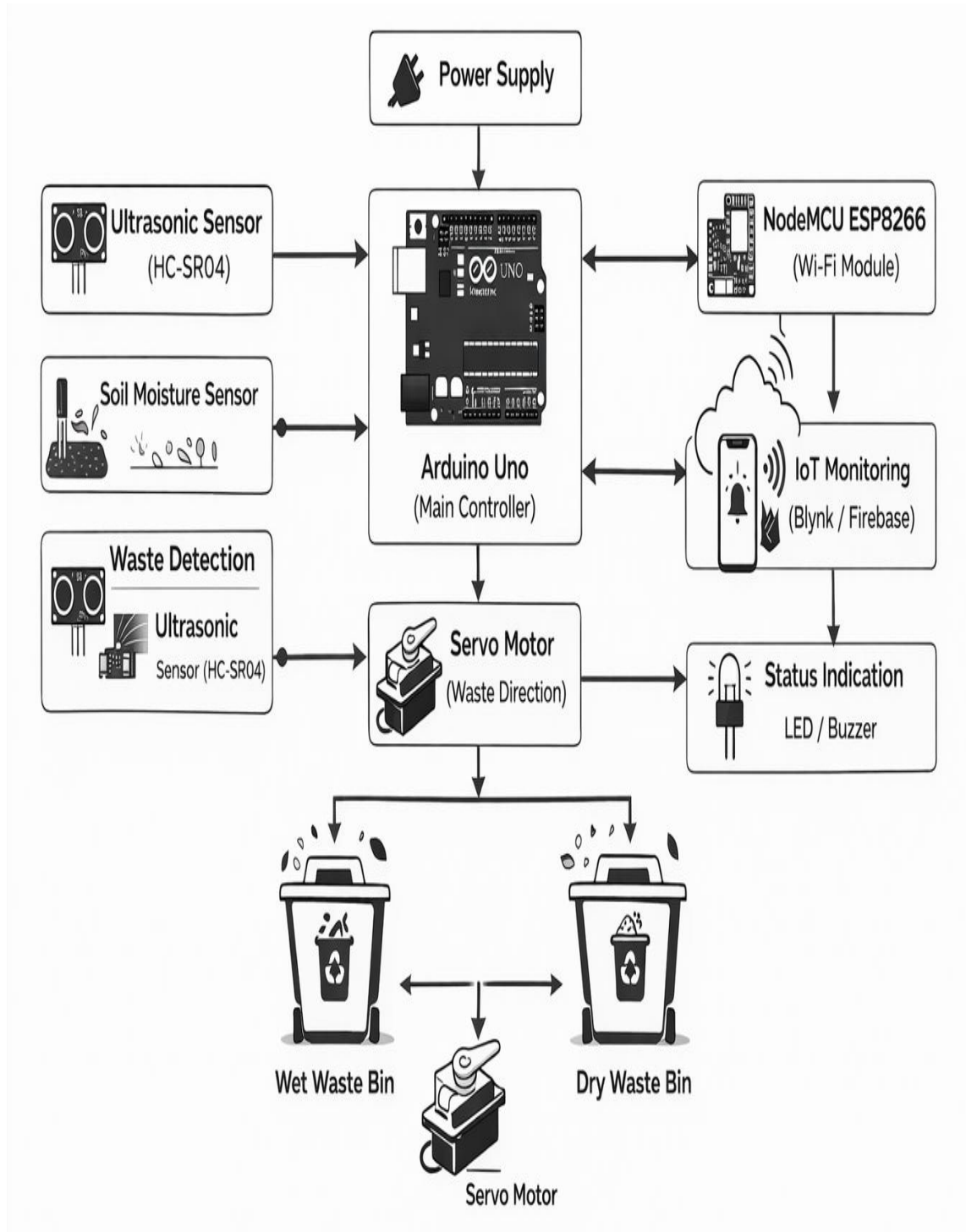
The proposed system aims to provide a simple, affordable, and eco-friendly solution suitable for household applications. By implementing automatic segregation at the initial stage, overall waste management efficiency can be significantly improved, reducing environmental impact and promoting sustainable waste handling practices.

LITERATURE SURVEY

Literature Review Table

Author	Year	Method/ Technology	Contribution	Limitation
Hoornweg & Bhada-Tata[1]	2012	Global waste management study	Highlighted importance of proper waste segregation for environmental protection	Focuses on analysis, not automation
Pushpa M.K. et al.[2]	2015	Microcontroller-based waste segregator	Developed automatic waste segregation using embedded systems and sensors	Limited classification capability
Vicentini et al.[3]	2008	Sensor-based waste container monitoring	Introduced smart bins that estimate waste levels for optimized collection	Requires advanced infrastructure
Mahajan Chitode[4]	& 2014	ZigBee-based waste bin monitoring	Enabled real-time wireless monitoring of garbage bins	Does not perform automatic segregation
Arebey al[5].	et 2011	IoT and sensor-based waste monitoring	Provided overview of smart waste management systems using sensors and communication networks	Mainly theoretical overview

BLOCK DIAGRAM



PROPOSED SYSTEM

The proposed system consists of:

- Arduino Uno microcontroller
- Ultrasonic sensor (HC-SR04)
- Soil moisture sensor
- Servo motor
- NodeMCU ESP8266
- Dual-bin structure (Dry & Wet)
- LED/Buzzer indication

Arduino Uno

The Arduino Uno acts as the central controller of the proposed waste segregation system. It is based on the ATmega328P microcontroller and is responsible for processing input signals and controlling output devices. The ultrasonic sensor and soil moisture sensor are connected to the input pins of the Arduino, while the servo motor is connected to a PWM output pin. When waste is detected by the ultrasonic sensor, the Arduino reads the moisture sensor value to determine whether the waste is wet or dry. Based on this analysis, it sends a control signal to rotate the servo motor toward the appropriate bin. The Arduino ensures smooth and accurate operation of the entire system.

Ultrasonic Sensor

The ultrasonic sensor (HC-SR04) is fixed at the inlet section to detect the presence of waste. The sensor works on the principle of sound wave reflection. It transmits ultrasonic pulses and receives the reflected echo signal. The time difference between transmission and reception is used to calculate the distance of the object from the sensor. When the measured distance falls below a predefined threshold, it indicates that waste has been placed into the system. This signal is then fed as input to the microcontroller to initiate the segregation process.

Soil Moisture Sensor

The soil moisture sensor is used to distinguish between wet and dry waste based on moisture content. The sensor consists of two conducting probes which measure the electrical conductivity of the material. Wet waste such as vegetable peels and leftover food contains high moisture and allows greater current flow between the probes. Dry waste such as paper and plastic contains minimal moisture and results in lower conductivity. The analog output from the

moisture sensor is continuously monitored by the microcontroller. If the sensor value exceeds the preset threshold, the waste is classified as wet; otherwise, it is classified as dry.

Servo Motor

The servo motor is used to control the direction of waste into the appropriate bin. Based on the classification result obtained from the moisture sensor, the microcontroller generates a control signal to rotate the servo motor to a specific angle. When wet waste is detected, the servo rotates towards the wet bin. When dry waste is detected, the servo rotates towards the dry bin. After directing the waste, the servo returns to its initial position, preparing the system for the next cycle.

NodeMCU ESP8266

The NodeMCU ESP8266 module is used for IoT-based monitoring of bin levels. An ultrasonic sensor is placed inside the bin to measure the fill level. The NodeMCU processes this data and sends it to the Blynk/Firebase platform through Wi-Fi.

When the bin becomes full, a notification is sent to the user, ensuring timely disposal and preventing overflow.

LED / Buzzer

An LED and buzzer are provided to give visual and audio indication of system operation. The indicator activates when waste is detected and classified, confirming proper functioning of the system.

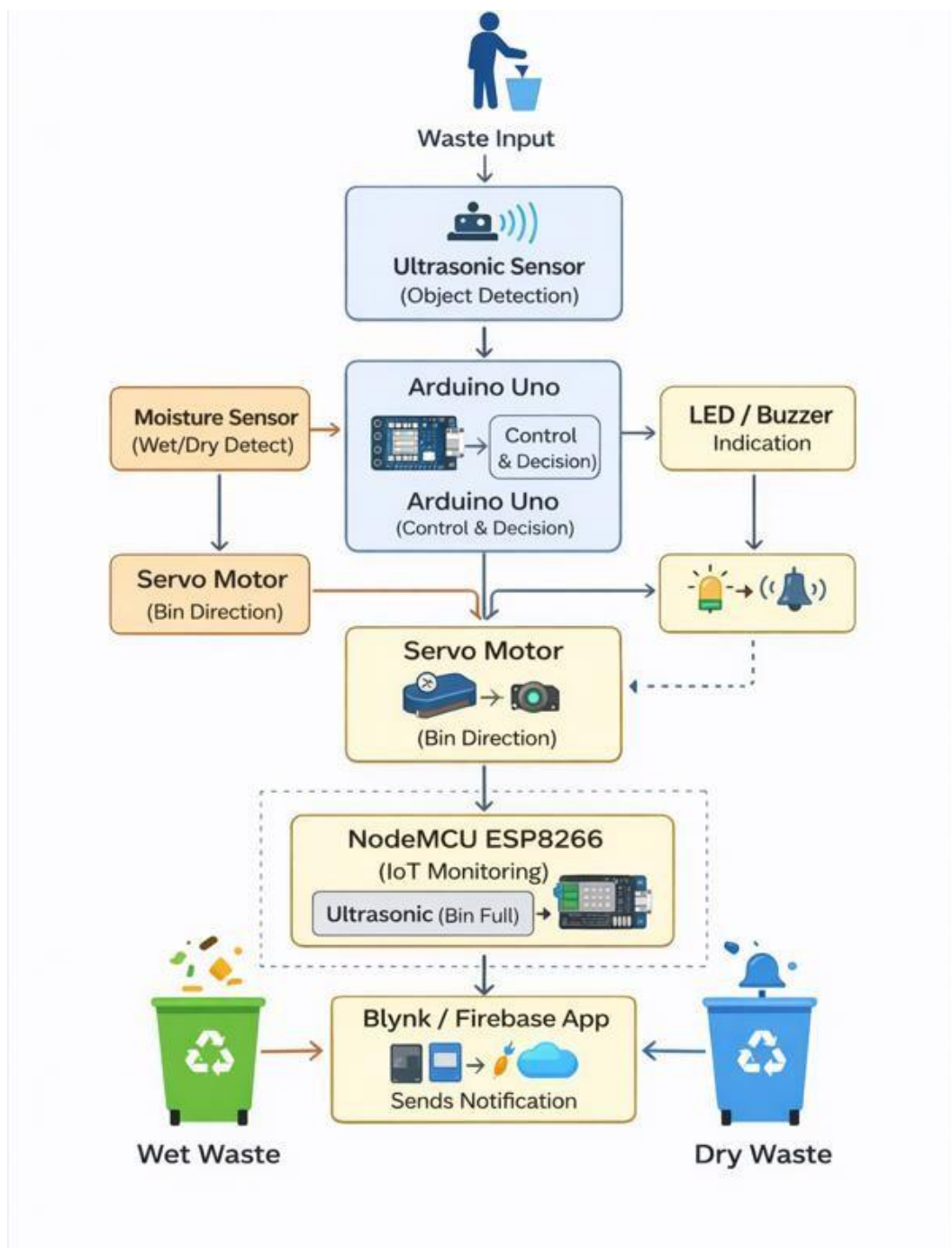


Fig. 1: Flowchart of Software Implementation

Software Implementation

The software implementation of the proposed system is carried out using the Arduino IDE and embedded C programming. The Arduino Uno is programmed to read inputs from the ultrasonic sensor and soil moisture sensor and to control the servo motor based on decision logic. The NodeMCU ESP8266 is programmed to monitor bin levels and send notifications through the IoT platform.

Initially, the system remains in idle mode. When waste is detected by the ultrasonic sensor, the Arduino reads the moisture sensor value and compares it with a predefined threshold. If the value exceeds the threshold, the waste is classified as wet; otherwise, it is classified as dry. Based on the result, the servo motor rotates toward the respective bin to deposit the waste.

Simultaneously, the NodeMCU continuously monitors the bin level using an ultrasonic sensor. When the waste level reaches the preset limit, a notification is sent to the user through the Blynk/Firebase application. The system then resets and waits for the next waste input.

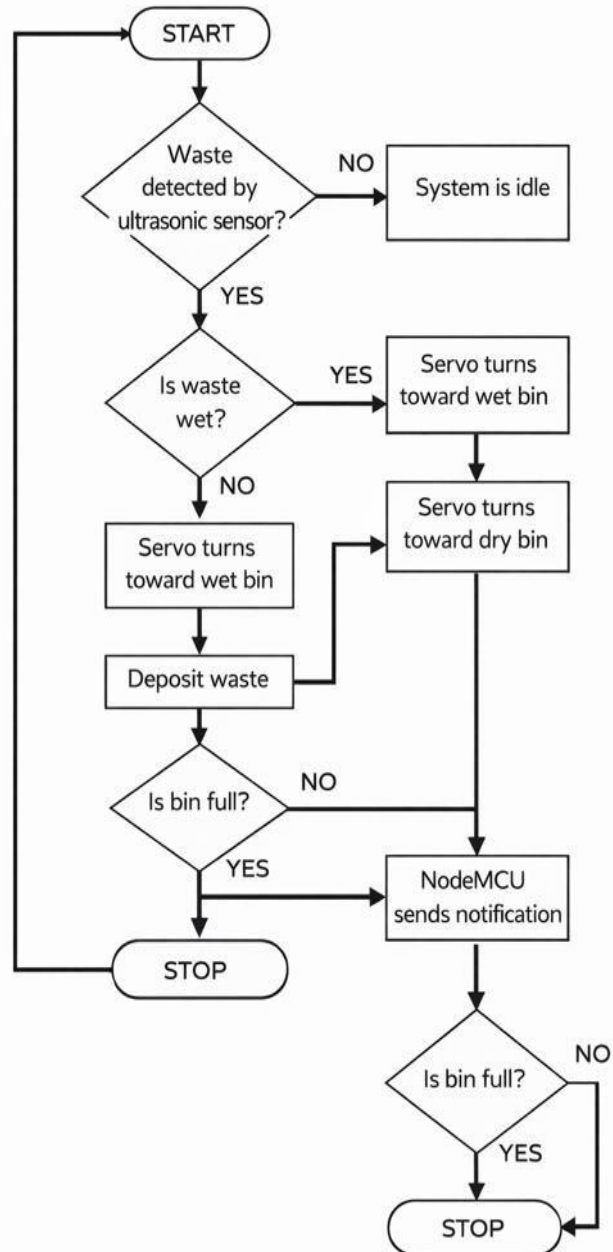


Fig. 2 : Flowchart of Software Implementation.

RESULTS AND DISCUSSION

The proposed Smart Dry and Wet Waste Segregation System was tested using common household waste materials such as vegetable peels, leftover food, paper, plastic bottles, and wrappers. The ultrasonic sensor successfully detected the presence of waste at the inlet, initiating the segregation process. The soil moisture sensor effectively differentiated wet and dry waste based on moisture content.

During testing, wet waste samples such as fruit peels and cooked food produced higher sensor values and were correctly directed to the wet bin. Dry waste materials such as paper and plastic showed lower moisture readings and were successfully directed to the dry bin. The servo motor responded accurately to the control signals generated by the Arduino Uno, ensuring proper waste direction.

Table 1 : Result of Wet Waste Detection

Sl. No.	Type of Metal Waste	Classified as Wet
1	Vegetable peel	Yes
2	Fruit waste	Yes
3	Leftover cooked food	Yes
4	Tea waste	Yes

Table 2: Result of Dry Waste Detection

Sl. No.	Types of Organic Waste	Classified as Dry
1	Paper	Yes
2	Plastic bottle	Yes
3	Wrapper	Yes
4	Cardboard	Yes

Table 3: Semi-Dry / Mixed Waste Testing.

Sl. No	Type of Waste	Classified Result
1	Damp paper	Wet
2	Moist cardboard	Wet
3	Half-dried food waste	Wet

The IoT monitoring unit using NodeMCU ESP8266 effectively measured the bin level through the ultrasonic sensor. When the bin reached the predefined threshold level, notifications were successfully sent via the Blynk/Firebase platform. This ensured timely waste disposal and prevented overflow.

However, minor variations in moisture readings were observed when semi-dry waste materials were tested. Proper calibration of the moisture sensor threshold improved classification accuracy. The system performed reliably under normal household conditions and demonstrated effective automation at the source level.

Overall, the experimental results indicate that the proposed system is capable of accurately segregating wet and dry waste while providing real-time monitoring, making it suitable for household waste management applications.

ACCURACY ANALYSIS

The classification accuracy was calculated based on the number of correctly identified samples.

- Total Samples Tested: 12
- Correctly Classified: 11
- Misclassified: 1

$$\text{Accuracy} = (11/12) \times 100 = 91.6\% \text{ (Approximately 92\%)}$$

FUTURE SCOPE

- Addition of metal detection sensor for multi-category segregation.
- Integration of AI and image processing for improved accuracy.
- Implementation of solar power supply for energy efficiency.
- Cloud-based data storage for waste analysis.
- Mobile application for centralized monitoring.
- Expansion for use in apartments, offices, and smart city projects.

CONCLUSION

- The proposed system successfully segregates wet and dry waste automatically.
- Moisture sensor accurately classifies waste based on moisture content.
- Arduino Uno controls the decision-making and servo operation efficiently.
- NodeMCU ESP8266 enables real-time bin level monitoring and notification.
- The system reduces manual effort and improves waste management efficiency.

- It is cost-effective, compact, and suitable for household applications.

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